DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
MONDAY TEST

CLASS: X **FULL MARKS: 40**
SUBJECT: PHYSICS **DATE: 30.09.2024**

General Instructions:

- This paper consists of <u>three</u> printed pages.
- All questions are <u>compulsory</u>.
- Marks will be deducted for spelling errors.

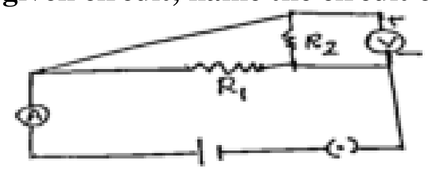
SECTION A

(Attempt all questions.)

Question 1

Choose the correct answers to the questions from the given options. [1 × 8 = 8]

(i) In the given circuit, name the circuit components that are connected in parallel.

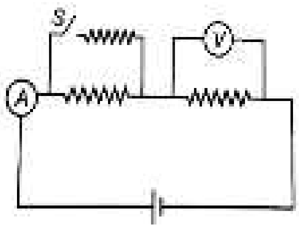


(a) R_1 and R_2 (b) R_1, R_2 and V (c) R_2 and V (d) R_1 and V

(ii) The main switch breaks the connection of
(a) phase wire and neutral wire (b) neutral wire and earth wire
(c) phase wire and earth wire (d) only phase wire

(iii) Bulb P is marked as 100 W, 220 V and bulb Q is marked as 60 W, 110 V. The ratio of resistance of P and Q is:
(a) 12: 7 b) 5: 7 c) 5: 12 d) 12: 5

(iv) Assertion (A): In the circuit shown in figure after closing the switch S, reading of ammeter will increase while that of voltmeter will decrease.
Reason (R): Net resistance decreases as closing the key makes a parallel combination of resistors.



(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

(v) A coil in the heater consumes power P on passing current. If it is cut into halves and joined in parallel, it will consume power
(a) P (b) P/2 (c) 2P (d) 4P

(vi) **Which of the following is not a characteristic of parallel combination of resistors?**

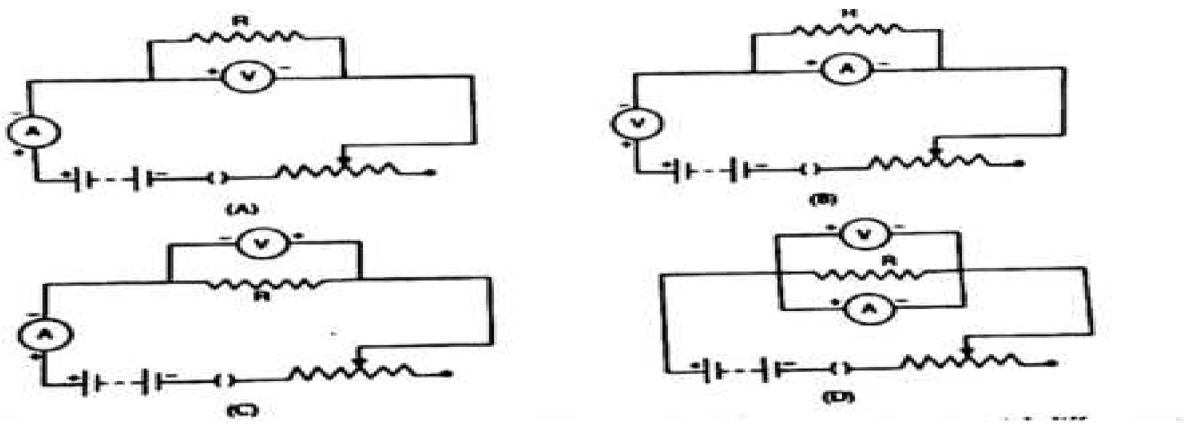
(a) If one resistor is fused, the circuit does not become open.

(b) The total resistance R is given by the formula $1/R = 1/R_1 + 1/R_2 + 1/R_3$ ……

(c) The total resistance becomes less than the least resistor, present in the combination.

(d) The current through each resistor always remains the same.

(vii) **Consider the following circuit diagrams drawn by four students:**



The correct circuit diagram for studying the dependence of current on the potential difference across the resistor is:

(a) A **(b) B** **(c) C** **(d) D**

(viii) **If three resistors R_1, R_2 and R_3 are connected in series and by $R_1 > R_2 > R_3$, then what is the relation between the potential drop V_1, V_2 and V_3 across R_1, R_2 and R_3 respectively?**

(a) $V_1 = V_2 = V_3$ **(b) $V_1 < V_2 < V_3$** **(c) $V_1 > V_2 > V_3$** **(d) $1/V_1 < 1/V_2 < 1/V_3$**

Question 2

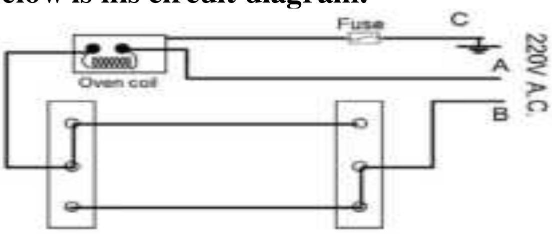
(i) Complete the following by choosing the correct answers from the bracket: **[3]**

(a)From the power generating station, before transmission of power the voltage is _____ *[stepped up/ stepped down/ maintained at the same value]*

(b) The current rating of a fuse connected to an electric appliance is _____ *[slightly higher/ slightly lower/equal to]* the maximum current that can be drawn by the appliance.

(c) The most appropriate material to make the filament of a bulb is ______________ *[manganin/tungsten/nichrome]*.

(ii) Purvi's friend Tim wants to connect a fuse to his oven. He wants to control the oven from two different locations. Shown below is his circuit diagram. **[2]**



(a) Which one of the two, A or B should be a live wire?

(b) In the event of an overload, will the fuse serve its purpose here?

Question 3

(i) (a) Name the physical quantity measured by $\sqrt{(H/Rt)}$ **[2]**

(b) Mention the SI unit of conductivity.

(ii) How does an increase in the temperature affect the conductance of a [2]
(a) Metal and (b) Semiconductor?

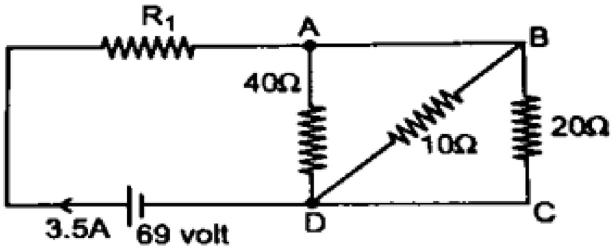
(iii) The electric meter installed in Trisha's house reads 210 units of electrical energy consumed at the end of November 2023.
(a) In which unit does the domestic electric meter measure the electrical energy consumed?
(b) Write a relation between the unit stated by you in part (a) above and the S.I. unit of energy.
(c) How much would she have to pay in order to clear her electricity bill if each unit costs Rs. 7.50? [3]

SECTION B

(Attempt all questions.)

Question 4 [3+3+4 = 10]

i) Calculate the resistance R_1 in the circuit shown.



ii) An electric bulb is marked 100 W, 230 V. If the supply voltage drops to 115 V, what is the heat energy produced by the bulb in 20 min?

iii) (a) Name the color codes of the wires that are connected to metallic body and switch of an appliance.
(b) Name a material that shows negligible increase in resistance with temperature.
(c) Mention one application of the material mentioned in the previous answer.

Question 5 [3+3+4 = 10]

i) There are two copper wires of length ratio 1:4 that have their cross-sectional areas in the ratio 1:2. What will be the ratio of their: (a) resistances? (b) specific resistances?

ii) While inserting a 3-pin plug in a socket, Trina found that the pins of the plug are of different dimensions. She was curious and asked her elder sister, who was studying in Class X, about the possible reason for her observations.
a) What are the reasons for different dimensions of the pins?
b) Why are different sizes of plugs available?

iii) In the below circuit diagram, calculate:
a) the resistance of the circuit.
b) the current I_2.
c) the current I.